CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge Ordinary Level

MARK SCHEME for the May/June 2015 series

4037 ADDITIONAL MATHEMATICS

4037/11 Paper 1, maximum raw mark 80

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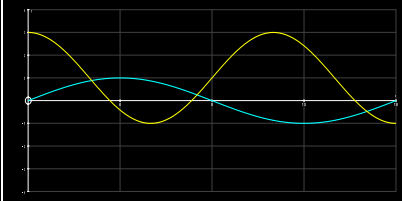
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Abbreviations

awrt answers which round to
cao correct answer only
dep dependent
FT follow through after error
isw ignore subsequent working
oe or equivalent
rot rounded or truncated
SC Special Case
soi seen or implied
www without wrong working

1	**(i)**	180° or π radians or 3.14 radians (or better)	**B1**	
	(ii)	2	**B1**	
	(iii) **(a)**		**B1**	$y = \sin 2x$ all correct
	(b)		**B1**	for either ↑↓↑starting at their highest value and ending at their lowest value Or a curve with highest value at $y = 3$ and lowest value at $y = -1$
			B1	completely correct graph
	(iv)	3	**B1**	
2	**(i)**	$\tan\theta = \dfrac{(8+5\sqrt{2})(4-3\sqrt{2})}{(4+3\sqrt{2})(4-3\sqrt{2})}$ $= \dfrac{32-24\sqrt{2}+20\sqrt{2}-30}{16-18}$ $= 1+2\sqrt{2}$ cao	**M1** **A1**	attempt to obtain $\tan\theta$ and rationalise. Must be convinced that no calculators are being used

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(ii)	$\sec^2\theta = 1+\tan^2\theta$ $= 1+\left(-1+2\sqrt{2}\right)^2$ $= 1+1-4\sqrt{2}+8$ $= 10-4\sqrt{2}$ **Alternative solution:** $AC^2 = \left(4+3\sqrt{2}\right)^2+\left(8+5\sqrt{2}\right)^2$ $= 148+104\sqrt{2}$ $\sec^2\theta = \dfrac{148+104\sqrt{2}}{\left(4+3\sqrt{2}\right)^2}$ $= \dfrac{148+104\sqrt{2}}{\left(4+3\sqrt{2}\right)^2}\times\dfrac{34-24\sqrt{2}}{34-24\sqrt{2}}$ $= 10-4\sqrt{2}$	**M1** **DM1** **A1** **M1** **DM1** **A1**	attempt to use $\sec^2\theta = 1+\tan^2\theta$, with *their* answer to **(i)** attempt to simplify, must be convinced no calculators are being used. Need to expand $\left(-1+2\sqrt{2}\right)^2$ as 3 terms
3 **(i)**	$64+192x^2+240x^4+160x^6$	**B3,2,1,0**	–1 each error
(ii)	$\left(64+192x^2+240x^4\right)\left(1-\dfrac{6}{x^2}+\dfrac{9}{x^4}\right)$ Terms needed $64-(192\times 6)+(240\times 9)$ $= 1072$	**B1** **M1** **A1**	expansion of $\left(1-\dfrac{3}{x^2}\right)^2$ attempt to obtain 2 or 3 terms using *their* **(i)**

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4	**(a)**	$\mathbf{X}^2 = \begin{pmatrix} 4-4k & -8 \\ 2k & -4k \end{pmatrix}$	**B2,1,0**	–1 each incorrect element
	(b)	Use of $\mathbf{AA}^{-1} = \mathbf{I}$	**M1**	use of $\mathbf{AA}^{-1} = \mathbf{I}$ and an attempt to obtain at least one equation.
		$\begin{pmatrix} a & 1 \\ b & 5 \end{pmatrix}\begin{pmatrix} \frac{5}{6} & -\frac{1}{6} \\ -\frac{2}{3} & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$		
		Any 2 equations will give $a = 2$, $b = 4$	**A1,A1**	
		Alternative method 1:		
		$\frac{1}{5a-b}\begin{pmatrix} 5 & -1 \\ b & a \end{pmatrix} = \begin{pmatrix} \frac{5}{6} & -\frac{1}{6} \\ -\frac{2}{3} & \frac{1}{3} \end{pmatrix}$	**M1**	correct attempt to obtain $\mathbf{A}^{-1}$ and comparison of at least one term.
		Compare any 2 terms to give $a = 2$, $b = 4$	**A1,A1**	
		Alternative method 2:		
		Inverse of $\frac{1}{6}\begin{pmatrix} 5 & -1 \\ -4 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 5 \end{pmatrix}$	**M1** **A1,A1**	reasoning and attempt at inverse
5		$3x - 1 = x(3x-1) + x^2 - 4$ or $y = \left(\frac{y+1}{3}\right)y + \left(\frac{y+1}{3}\right)^2 - 4$		
		$4x^2 - 4x - 3 = 0$ or $4y^2 - 4y - 35 = 0$ $(2x-3)(2x+1) = 0$ or $(2y-7)(2y+5) = 0$	**M1** **DM1**	equate and attempt to obtain an equation in 1 variable forming a 3 term quadratic equation and attempt to solve
		leading to $x = \frac{3}{2}, x = -\frac{1}{2}$ and	**A1**	x values
		$y = \frac{7}{2}, y = -\frac{5}{2}$	**A1**	y values
		Midpoint $\left(\frac{1}{2}, \frac{1}{2}\right)$	**B1**	for midpoint, allow anywhere
		Perpendicular gradient = $-\frac{1}{3}$	**M1**	correct attempt to obtain the gradient of the perpendicular, using AB
		Perp bisector: $y - \frac{1}{2} = -\frac{1}{3}\left(x - \frac{1}{2}\right)$	**M1**	straight line equation through the midpoint; must be convinced it is a perpendicular gradient.
		$(3y + x - 2 = 0)$	**A1**	allow unsimplified

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6 (i)	$f\left(\frac{1}{2}\right)=\frac{a}{8}-\frac{15}{4}+\frac{b}{2}-2=0$ leading to $a+4b=46$ $f(1)=a-15+b-2=5$ leading to $a+b=22$ giving $b=8$ (AG), $a=14$	**M1** **A1** **M1,A1**	correct use of either $f\left(\frac{1}{2}\right)$ or f(1) paired correctly both equations correct (allow unsimplified) M1 for solution of equations A1 for both a and b. **AG** for b.
(ii)	$(2x-1)(7x^2-4x+2)$	**M1,A1**	M1 for valid attempt to obtain g(x), by either observation or by algebraic long division.
(iii)	$7x^2-4x+2=0$ has no real solutions as $b^2<4ac$ $16<56$	**M1** **A1**	use of b^2-4ac correct conclusion; must be from a correct g(x) or 2g(x) www
7 (i)	$\frac{dy}{dx}=\frac{(x-1)\frac{8x}{(4x^2+2)}-\ln(4x^2+3)}{(x-1)^2}$ When $x=0$, $y=-\ln 3$ oe $\frac{dy}{dx}=-\ln 3$ so gradient of normal is $\frac{1}{\ln 3}$ (allow numerical equivalent) normal equation $y+\ln 3=\frac{1}{\ln 3}x$ or $y=0.910x-1.10$, or $y=\frac{10}{11}x-\frac{11}{10}$ cao (Allow $y=0.91x-1.1$)	**M1** **B1** **A1** **B1** **M1** **M1** **A1**	differentiation of a quotient (or product) correct differentiation of $\ln(4x^2+3)$ all else correct for y value valid attempt to obtain gradient of the normal attempt at normal equation must be using a perpendicular
(ii)	when $x=0$, $y=-\ln 3$ when $y=0$, $x=(\ln 3)^2$ Area = ±0.66 or ±0.67 or awrt these or $\frac{1}{2}(\ln 3)^3$	**M1** **A1**	valid attempt at area

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8	(i)	Range for f: $y \geq 3$	B1	
		Range for g: $y \geq 9$	B1	
	(ii)	$x = -2 + \sqrt{y-5}$	M1	attempt to obtain the inverse function
		$g^{-1}(x) = -2 + \sqrt{x-5}$	A1	Must be correct form
		Domain of g^{-1}: $x \geq 9$	B1	for domain
		Alternative method:		
		$y^2 + 4y + 9 - x = 0$	M1	attempt to use quadratic formula and find inverse
		$y = \dfrac{-4 + \sqrt{16 - 4(9-x)}}{2}$	A1	must have + not ±
	(iii)	Need $g(3e^{2x})$	M1	correct order
		$(3e^{2x} + 2)^2 + 5 = 41$	DM1	correct attempt to solve the equation
		or $9e^{4x} + 12e^{2x} - 32 = 0$		
		$(3e^{2x} - 4)(3e^{2x} + 8) = 0$		
		leading to $3e^{2x} + 2 = \pm 6$ so $x = \frac{1}{2}\ln\frac{4}{3}$	M1	dealing with the exponential correctly in order to reach a solution for x
		or $e^{2x} = \frac{4}{3}$ so $x = \frac{1}{2}\ln\frac{4}{3}$	A1	Allow equivalent logarithmic forms
		Alternative method:		
		Using $f(x) = g^{-1}(41)$, $g^{-1}(41) = 4$	M1	correct use of g^{-1}
		leading to $3e^{2x} = 4$, so $x = \frac{1}{2}\ln\frac{4}{3}$	DM1	dealing with $g^{-1}(41)$ to obtain an equation in terms of e^{2x}
			M1	dealing with the exponential correctly
			A1	in order to reach a solution for x Allow equivalent logarithmic forms
	(iv)	$g'(x) = 6e^{2x}$	B1	B1 for each
		$g'(\ln 4) = 96$	B1	

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9 (i)	$\frac{dy}{dx} = 3x^2 - 10x + 3$	**M1**	for differentiation
	When $x = 0$, for curve $\frac{dy}{dx} = 3$, gradient of line also 3 so line is a tangent.	**A1**	comparing both gradients
	Alternate method:		
	$3x + 10 = x^3 - 5x^2 + 3x + 10$	**M1**	attempt to deal with simultaneous equations
	leading to $x^2 = 0$, so tangent at $x = 0$	**A1**	obtaining $x = 0$
(ii)	When $\frac{dy}{dx} = 0$, $(3x-1)(x-3) = 0$	**M1**	equating gradient to zero and valid attempt to solve
	$x = \frac{1}{3}$, $x = 3$	**A1,A1**	A1 for each
(iii)	Area $= \frac{1}{2}(10+19)3 - \int_0^3 x^3 - 5x^2 + 3x + 10\,dx$	**B1**	area of the trapezium
	$= \frac{87}{2} - \left[\frac{x^4}{4} - \frac{5x^3}{3} + \frac{3x^2}{2} + 10x\right]_0^3$	**M1**	attempt to obtain the area enclosed by the curve and the coordinate axes, by integration
	$= \frac{87}{2} - \left(\frac{81}{4} - 45 + \frac{27}{2} + 30\right)$	**A1**	integration all correct
		DM1	correct application of limits (must be using *their* 3 from **(ii)** and 0)
	$= 24.7$ or 24.8	**A1**	
	Alternative method:		
	Area $= \int_0^3 (3x+10) - \left(x^3 - 5x^2 + 3x + 10\right)dx$	**B1**	correct use of '*Y*–*y*'
	$= \int_0^3 -x^3 + 5x^2\,dx$	**M1**	attempt to integrate
		A1	integration all correct
	$= \left[-\frac{x^4}{4} + \frac{5x^3}{3}\right]_0^3 = \frac{99}{4}$	**DM1**	correct application of limits
		A1	
10 (a)	$\sin^2 x = \frac{1}{4}$		
	$\sin x = (\pm)\frac{1}{2}$	**M1**	using $\text{cosec}\, x = \frac{1}{\sin x}$ and obtaining $\sin x = \ldots$
	$x = 30°, 150°, 210°, 330°$	**A1,A1**	A1 for one correct pair, A1 for another correct pair with no extra solutions

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(b)	$(\sec^2 3y - 1) - 2\sec 3y - 2 = 0$	**M1**	use of the correct identity
	$\sec^2 3y - 2\sec 3y - 3 = 0$ $(\sec 3y + 1)(\sec 3y - 3) = 0$ leading to $\cos 3y = -1$, $\cos 3y = \frac{1}{3}$	**M1**	attempt to obtain a 3 term quadratic equation in sec 3y and attempt to solve dealing with sec and 3y correctly
	$3y = 180°, 540°$ $\quad 3y = 70.5°, 289.5°, 430.5°$ $y = 60°, 180°, 23.5°, 96.5°, 143.5°$	**A1,A1** **A1**	A1 for a correct pair, A1 for a second correct pair, A1 for correct 5th solution and no other within the range
	Alternative 1:		
	$\sec^2 3y - 2\sec 3y - 3 = 0$	**M1**	use of the correct identity
	leading to $3\cos^2 3y + 2\cos 3y - 1$ $(3\cos y - 1)(\cos y + 1) = 0$	**M1** **M1**	attempt to obtain a quadratic equation in cos 3y and attempt to solve dealing with 3y correctly **A** marks as above
	Alternative 2:		
	$\frac{\sin^2 y}{\cos^2 y} - \frac{2}{\cos y} - 2 = 0$ $(1 - \cos^2 x) - 2\cos x - 2\cos^2 x = 0$	**M1**	use of the correct identity, $\tan y = \frac{\sin y}{\cos y}$ and $\sec y = \frac{1}{\cos y}$, then as before
(c)	$z - \frac{\pi}{3} = \frac{\pi}{3}, \frac{4\pi}{3}$	**M1**	correct order of operations
	$z = \frac{2\pi}{3}, \frac{5\pi}{3}$ or 2.09 or 2.1, 5.24	**A1,A1**	A1 for a correct solution A1 for a second correct solution and no other within the range

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